Shanmugam Subramanian

Mechanical Engineer II, Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience in antenna and radar hardware. Owned eight 5G macro and DAS products with a >90% FAI pass rate. Skilled in FEA, CFD, RCA, risk mitigation, and feasibility analysis.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

Product Development & NPI

- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna System products for the US & Canada markets.
- Released RF PCB, sheet metal, die-cast, CNC, & EDM parts with a >**90% FAI pass rate**.
- Managed **ECRs, FAIs, & SVPRs** using Salesforce Propel **PLM** & SolidWorks **PDM** in coordination with the global team.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.
- Built RF PCB fixtures, press-pull tools, & alignment jigs: **40% less rework, 75% faster riveting**, 20–50 min saved per unit.

Mechanical Design

- Designed pole, wall, ceiling, & tri-sector **mounting brackets**; gravity-assisted design **halved installation manpower**.
- Drove **simulation-led design** (snap-fit, topology, FEA) for 3 molded parts & 1 FRP radome: **first-pass success**, ∼**$15K under budget**.
- Introduced **topology-optimized** FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.

CAE, Simulation & CFD

- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Ran **non-linear FEA** of a production snap-fit: predicted 38 N insertion & 30 N removal force; flagged a fillet redesign.
- **Ansys Fluent CFD** radome study (2D/3D): DOE on 135+ profiles gave **12–18% wind load reduction**; **patent** in progress.
- Built **Excel hand-calculation tools** for drag force, pneumatic crimping force, & plastic snap-joint behavior.
- Automated **Ansys Mechanical pre-processing** with Python/ACT scripts (coarse/medium/fine meshing, material assignment, midsurfacing thin-walled bodies, BC & solver setup), **cutting setup time** ∼**2 hrs per simulation job** & standardizing models.
- Performed **IEC 60068-2-31 drop tests** for mechanical qualification of antenna products.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Developed two automotive radar antenna chassis, & radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Enhanced chassis's thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Lowered radome **cost by 8%** through material optimization without degrading dielectric performance.
- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.
- Created company-wide central library in SolidWorks with 700+ frequently used fasteners, reducing design lead time.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

Mini Trencher Design — APRIL/ MAY - 2021 3rd Year Project B.E.
Designed a low-cost mini agricultural trencher; reduced 5× cost & size through drivetrain & structure optimization.

Prodigy Design Competition— May 2021 NIT Trichy
Designed a rolling-belt mechanism for water-body waste removal in 24 hours; runner-up among 50 teams.

6-DOF Mimicking Robot— July 2025- Nov 2025 Personal project
Engineered a 6-DOF robotic arm with real-time mirroring, trajectory recording, and repeatable playback of arm movements.

Gearbox design — Dec 2025 Personal project
Designed and built a 1:64 multi-stage spur gear gearbox system to power a conveyor platform for scaled vehicle testing.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation& Motion Analysis
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, C, C++, Arduino
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Management & Quality: RCA, Risk Mitigation, Feasibility Analysis, Cross-functional Team Coordination